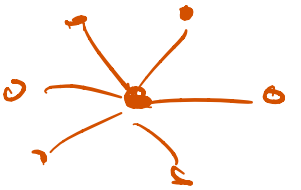
Recall $\deg v$ = number of edges incident with v, $\delta(G) = \min\{\deg v : v \in V(G)\}$, $\Delta(G) = \max\{\deg v : v \in V(G)\}$.
Goal: relations between vertex degrees and connectivity.

Example. *If a graph G in n vertices contains a vertex of degree $n-1$, then G is connected and $\operatorname{diam}(G) \leqslant 2$.*

If $\deg v = n-1$, u, w vertices, Then (u, v, w) is u–w walk.



Theorem (2.4). *Let G be a graph in n vertices. If*

$$\deg u + \deg v \geqslant n - 1$$

for every two nonadjacent vertices u and v, then G is connected and $\operatorname{diam}(G) \leqslant 2$.

Proof: WTS: if u, v are vertices, Then $\exists$ u-v path of length ≤ 2

Case 1: $uv \in E(G)$, Then we have a path of length 1 ✓

Case 2: $uv \notin E(G)$. Let's see There is $w \in N(u) \cap N(v)$. Otherwise

$$|N(u) \cup N(v)| = |N(u)| + |N(v)| - |N(u) \cap N(v)|$$

$$|N(u) \cup N(v)| = \deg u + \deg v$$

But $n-2 \geq |N(u) \cup N(v)|$ because $u, v \notin N(u) \cup N(v)$

and $\deg u + \deg v \geq n-1$ by assumption

Then $n-2 \geq |N(u) \cup N(v)| = \deg u + \deg v \geq n-1$, contradiction

Hence there is $w \in N(u) \cap N(v)$ and (u, w, v) is a u-v path of length 2. ▨

Note: The bound $n-1$ is sharp, i.e., there is a disconnected graph with $\deg u + \deg v \geqslant n-2$ for every two nonadjacent vertices u and v.



$n = 2m$

Corollary (2.5). *If G is a graph in n vertices and $\delta(G) \geqslant \frac{n-1}{2}$, then G is connected.*

Proof: WTS: if u, v are nonadjacent, Then $\deg u + \deg v \geqslant n-1$

But $\deg u + \deg v \geqslant \delta(G) + \delta(G) \geqslant \frac{n-1}{2} + \frac{n-1}{2} = n-1$,
By the previous Theorem, G is connected. ∎

Exercise (Exercise (2.11)). *Is the bound $\frac{n-1}{2}$ in the previous corollary sharp?*

Exercise (Exercise (2.9)). *Show that if G is a disconnected graph with exactly two odd vertices, then these odd vertices must be in the same component.* Hand shaking lemma.

Exercise (Exercise (2.7)). *Show that if G is a bipartite graph with partite sets X and Y, then $|E(G)| = \sum_{u \in X} \deg u = \sum_{v \in Y} \deg Y$. Can you generalize this statement for multipartite graphs?*

Definition. *A graph* G *is* ***regular*** *if all vertices have the same degree. If* $\deg v = r$ *for all* v*, we say that* G *is* ***r-regular.***

- Recall that $0 \leqslant \delta(G) \leqslant \Delta(G) \leqslant n-1$, for $n = |V(G)|$. Thus G is r-regular iff $\delta(G) = \Delta(G) = r$, where $r \in \{0, 1, \ldots, n-1\}$.

Example (All regular graphs in 4 and 5 vertices).

Order 4:

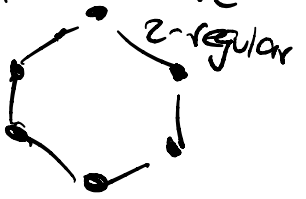








Cycles are 2-regular



Order 5:

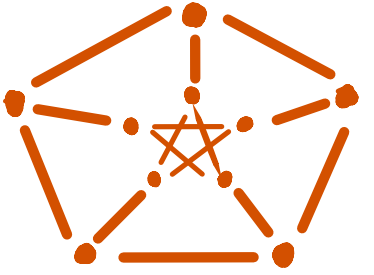






K_5

Example (The Petersen graph).



No 1-regular graph in 5-vertices

3-reg in 5-vertices,

Note: If n and r are both odd, there are no r-regular graphs in n-vertices (Follows from Corollary 2.3: every graph has an even number of odd vertices).

Next: there are no further restrictions for the existence of regular graphs. $r \leq n-1$

Notation: If $V(G) = \{v_1, \ldots, v_n\}$ we perform arithmetic in the subscripts modulo n.
E.g., if $n = 6$ and $i = 5$, then v_{i+2} denotes v_1.

Theorem (2.6, Harary graphs). *Fix integers r and n with $0 \leqslant r \leqslant n-1$. There exists an r-regular graph in n-vertices if and only if at least one of r and n is even.*

Proof: $\Rightarrow$) Suppose G is r-regular in n-vertices.

$$2|E(G)| = \sum_{v \in G} \deg v = nr \Rightarrow n \text{ or } r \text{ is even}$$

$\Leftarrow$) Consider G with $V(G) = \{v_1, \dots, v_n\}$.

Case 1: r is even, say $r = 2k$. Define edges:

$$E(G) = \left\{ v_i v_j : 1 \leqslant |i-j| \leqslant \frac{r}{2} = k \right\}$$

$$N(v_i) = \{ \underbrace{v_{i-k}, v_{i-k+1}, \dots, v_{i-1}}_{k \text{ vertices that precede } v}, \underbrace{v_{i+1}, \dots, v_{k-1}, v_k}_{k \text{ vertices that follow } v_k} \}$$

Contains $2k = r$ vertices

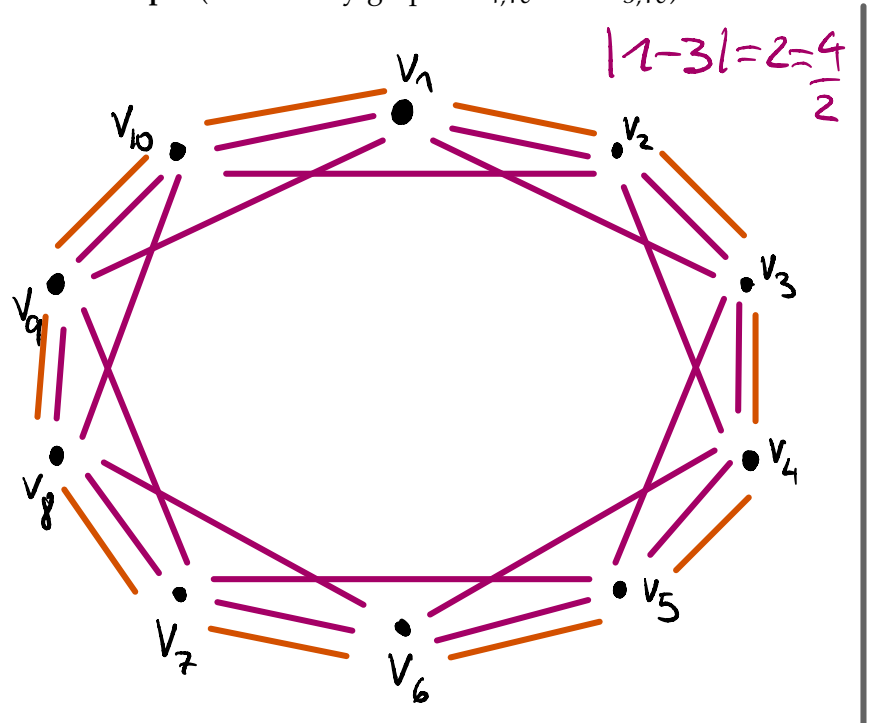
Case 2: $r = 2k+1$ odd, $n = 2m$ even.

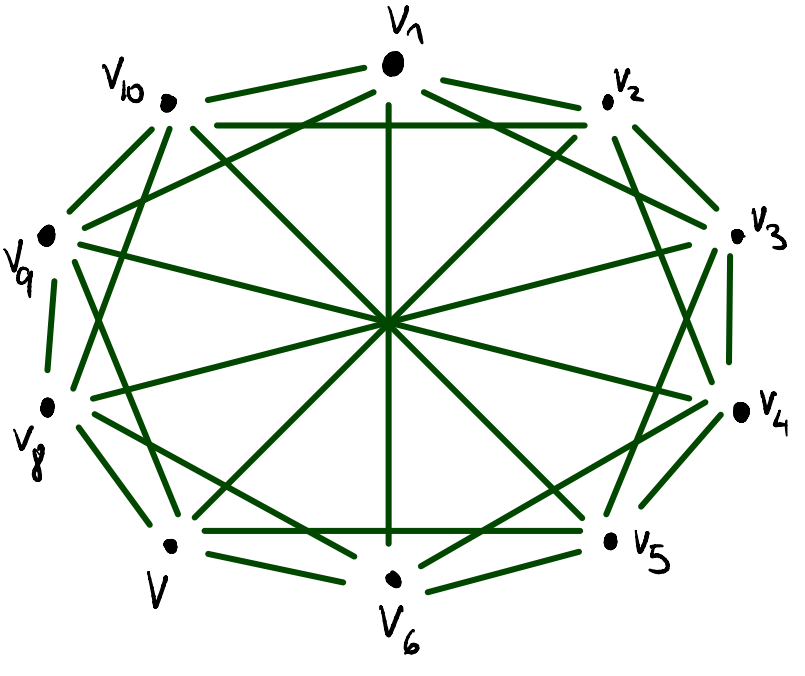
$$E(G) = \{ v_i v_j : 1 \le |i-j| \le \frac{r-1}{2} = k \} \cup \{ v_i v_{i+m} \}$$

$$N(v_i) = \{ \underbrace{v_{i-k}, v_{i-k+1}, \ldots, v_{i-1}}_{k \text{ vertices that precede } v}, \underbrace{v_{i+1}, \ldots, v_{k-1}, v_k}_{k \text{ vertices that follow } v_k} \} \cup \{ v_{i+m} \}$$

Contains $k + k + 1 = 2k+1$ vertices

Example (The Harary graphs $H_{4,10}$ and $H_{5,10}$).





Theorem (2.7). *For every* G *and every integer* $r \geqslant \Delta(G)$, *there exists an* r*-regular graph* H *containing* G *as an induced subgraph.*

Proof: Say $V(G) = \{v_1, \dots, v_n\}$.

Let G' the graph with $V(G) = \{v_1', \dots, v_n'\}$

with the same edges of G. $(v_i'v_j' \in E(G') \iff v_iv_j \in E(G))$

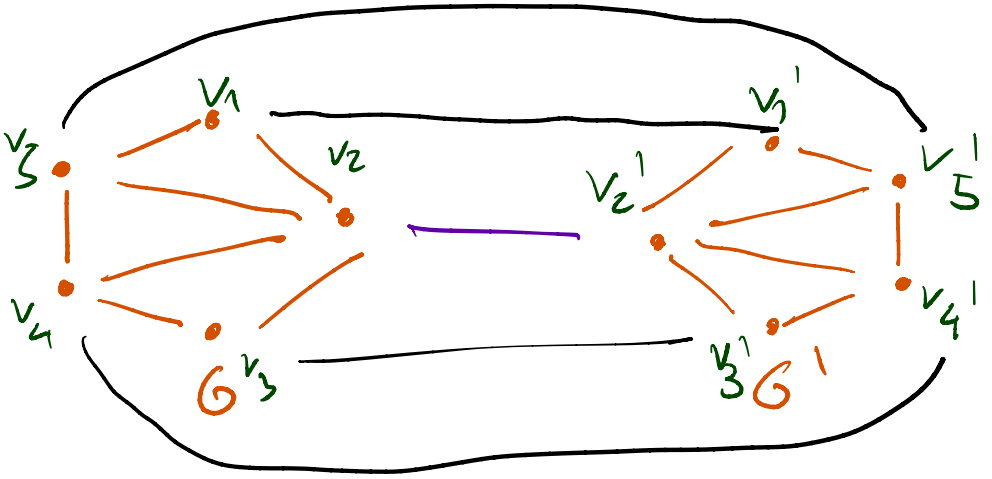
Consider $G_1 = G \cup G' + \{v_iv_i' \text{ if } \deg v_i < r\}$

Key: 1) $\delta(G_1) = \delta(G) + 1 \leq r$ 2) $G \subseteq G'$ induced

If G' is regular, STOP. Otherwise, continue procedure

which will give a regular graph G^k in at most $k = r - \delta(G)$ iterations

Example (An illustration of the proof of Theorem 2.7)**.**



$r = 4$

$r \geq 5$